Unit 3.1 Printables Teacher Key: Changing Earth, Changing Life

Standard: HS-ESS2-7. Construct an argument based on evidence about the simultaneous coevolution of Earth's systems and life on Earth. Targets are labeled ESS2-7.1 to ESS2-7.6.

Data caveats

• Dates and percentages from the geologic record are rounded and are teaching simplifications. Sources are cited on each item. Scientists revise ranges as new evidence arrives.

• The oxygen graph values are a teaching model built from published reconstructions (Lyons et al. 2014; Berner 2009). Estimates of past oxygen vary a lot; the peak near 30% is model-based.

• The "Big Five" extinction percentages are rounded approximations (Barnosky et al. 2011) and different studies give different ranges.

• Evidence cards (E1 to E6, C1 to C8) summarize types of evidence for classroom use; they are not quotations from the cited papers.

• Sorting by E, L or B has more than one defensible answer for several cards; accept reasoned answers.

How to use this key

Items are in lesson order and match the Printables index. Sample answers are model answers, not the only acceptable wording: accept any answer that uses the evidence correctly. Notebook warm-ups and exit tickets are listed under each lesson.

Lesson 1: Rusty Stripes

Targets: ESS2-7.1: I can name examples of how Earth has changed and how life has changed.

Card sort: 12 events in Earth and life history

Cards are the same 12 events listed on the printable. Answers for E, L or B are on the "Timeline strip and E / L / B chart" key.

Timeline strip and E / L / B chart

Card	E, L or B	Box	Why
A	E	1	Earth formed (no life yet)
B	E	3	Oceans form
C	L	6	Life begins
D	L	10	Photosynthesis spreads
E	B	11	Life made oxygen (L to E), which changed air and ocean chemistry
F	L	20	First animals; life changes
G	B	21	Plants change land and soil (L to E) and need land to live on
H	E	22	Pangaea: Earth change
I	B	22	Earth event (volcanoes) caused extinction of life; Earth conditions changed too
J	B	23	Asteroid (E) then extinctions (L)
K	B	23	Climate change affects life; life affects carbon dioxide
L	L	23	A new living thing appears

Accept any reasoned answer for E, L or B. Cards in boxes 20 to 23 (F to L) crowd together: more events are known from recent history because the evidence is better preserved, and life and Earth changed fast late in the record.

Unit CER evidence log

Rows will fill in over the unit. Example rows: Banded iron formation, rock, oxygen reached the oceans (about 2.4 billion years ago). Stromatolites, fossil, photosynthetic life existed early (3.5 billion years ago). Oxygen data, data, oxygen in the air rose over time.

Notebook sample answers

Pre-check (ungraded; compare at the end of the unit): Earth is about 4.5 billion years old; examples vary; living things did change air (photosynthesis) and rocks (soil, reefs).

Warm-up: rust needs oxygen, so oxygen reached the water that touched the rock; questions vary.

Exit ticket: e.g., Earth changed when oceans formed; life changed when photosynthesis spread.

Lesson 2: A Two-Way Street

Targets: ESS2-7.2: I can describe one way Earth changed life and one way life changed Earth.

Cause-and-effect cards

Answers are on the sorting chart key.

Sorting chart and diagrams

Card	Direction	What changed
1	L->E	Oxygen enters the air and oceans
2	E->L	Ash blocks sunlight and plants die
3	L->E	Rock becomes soil
4	E->L	Habitat is lost for animals
5	L->E	Reefs change the shoreline
6	E->L	Climate changes and animals evolve thick fur
7	L->E	Carbon dioxide is stored as carbon
8	E->L	Dust darkens the sky and species go extinct

4 are E->L and 4 are L->E. Change between Earth and life goes in **both** directions. Diagrams vary: look for two labeled boxes, an arrow, and a "First, ... so ..." sentence.

Notebook sample answers

Warm-up: ice sheet: plants and animals die or move; lichen: breaks the rock into tiny pieces and makes soil.

Find it in the reading: any sentence from pp. 219-225 showing life changing Earth (e.g., cyanobacteria released oxygen).

Exit ticket: Earth changed life: an ice age made animals evolve thick fur. Life changed Earth: lichens made soil.

Lesson 3: The Oxygen Story

Targets: ESS2-7.3: I can explain how events in Earth's history changed life, and how life in turn changed Earth.

Oxygen story: data table and graph grid

Oxygen first rises above a trace between about 2.5 and 2.0 billion years ago (label near 2.4 billion years ago, the Great Oxidation Event). It was made by photosynthesis by cyanobacteria. Oxygen is highest (about 30%) around 0.3 billion years ago, higher than today (21%); afterward it fell (model-based estimates; students might say it was used up or changed as forests were buried and climates shifted).

Points: (4.0, 0), (3.0, 0), (2.5, 0), (2.0, 2), (1.0, 3), (0.5, 15), (0.3, 30), (0, 21).

Mass extinction data and the rust record

The oxygen from living things came first: the iron could only rust after oxygen reached the water. This is life changing Earth (L->E): photosynthetic cells changed ocean and air chemistry.

End-Permian killed the greatest percent (about 96%). Earth events (eruptions, an asteroid) change climate, light, or oceans, so many species cannot survive; survivors spread into empty habitats and new groups evolve.

Build a chain

Chain: Cells use photosynthesis (L) -> **release oxygen** -> iron in the ocean **rusts and sinks** -> oxygen builds up in the air -> an ozone layer forms and **life can live on land**. Arrow marks: photosynthesis to oxygen L->E; oxygen to rusting E (chemical) shown as part of L->E; ozone to life on land E->L.

Sample sentence: Because cells began using photosynthesis, the ocean and air changed, which led to the ozone layer and life on land.

Notebook sample answers

Warm-up: there was little or no oxygen when the stripes formed because rust needs oxygen that arrived later.

Wrap-up: Because cells began using photosynthesis, the ocean and air changed, which led to the ozone layer and animal life on land.

Lesson 4: Three Case Files

Targets: ESS2-7.3: I can explain how events in Earth's history changed life, and how life in turn changed Earth.

Three case files

Case	Event	Effect on life	Effect back on Earth	Arrows
A	Oxygen from photosynthesis rises	Ozone lets life move onto land	Ozone layer forms; oxygen builds up	L->E then E->L
B	Plants spread on land	Larger plants grow in deep soil	Soil forms; carbon buried as coal; CO■ falls; climate cools	L->E, then E->L, then L->E
C	Huge eruptions release gases	About 96% of ocean species die; survivors spread; new groups evolve	Warmer climate; oceans lose oxygen	E->L, then life refills Earth

In common: each case is a chain of changes where an event in one system (air, rock, or life) causes changes in another, and the effect comes back.

Notebook sample answers

Quick write: e.g., Because plants spread on land, soil formed, which let larger plants grow. Because forests buried carbon, carbon dioxide fell, which cooled the climate.

Exit ticket: loop sentences vary.

Lesson 5: Make the Argument

Targets: ESS2-7.4: I can write an argument, with evidence, that Earth and life changed each other.

Evidence bank: six cards

Any three cards are acceptable for the evidence log. Kinds: E1 rock; E2 fossil; E3 data; E4 fossil and rock; E5 data; E6 rock.

Argument planner and peer-check scale

Sample plan: Claim: Earth and life changed each other because living things changed the air and the changed air changed living things. Evidence 1 (L->E): Banded iron formations about 2.4 billion years old show oxygen reached the ocean (rock). Evidence 2 (E->L): Five mass extinctions, such as the end-Permian, killed up to 96% of species after Earth events (data). Evidence 3: Oxygen rose from about 0% to 15% by 0.5 billion years ago, and the first animals appeared (data and fossil). Reasoning: Rust needs oxygen, so living things that make oxygen must have changed the ocean and air. Counterclaim: Volcanoes made the oxygen. That does not fit because stromatolite fossils show oxygen-making life existed.

Peer-check scores vary. A 4 needs a claim that goes both ways, three pieces of evidence with numbers from two or more kinds, reasoning, and a counterclaim.

Notebook sample answers

Warm-up: disagree; life changed Earth by making oxygen.

Argument paragraph: see the argument planner key.

Lesson 6: Evidence Court

Targets: ESS2-7.5: I can combine several kinds of evidence into one strong argument about Earth and life.

Evidence Court: eight evidence cards

Ratings are on the rating table key.

Rating table and argument frames

Card	Kind	Strength and why (sample)
C1	Rock	3: direct, checkable, agrees with C2 and C7
C2	Chemical data	3: measured and repeatable
C3	Fossil	3: direct evidence of early photosynthetic life
C4	Model	2: useful but depends on assumptions; strong only if it matches rock data
C5	Fossil	2: shows timing; does not show cause alone
C6	Rock	3: direct evidence of soil under forests
C7	Chemical data	3: measured signature of life
C8	Story	1: no data and cannot be checked

Sample argument: Claim: Life changed Earth's air and oceans. Evidence: banded iron formations (rock) show oxygen reached the oceans; stromatolites (fossil) show photosynthetic life 3.5 billion years ago; ancient rock chemistry (data) shows oxygen was nearly absent before 2.4 billion years ago. They agree because rust needs oxygen and living cells make it. Skeptic: the air was always the same (C8). That has no data, while the rocks do.

Notebook sample answers

Warm-up: no; more witnesses that agree make a story more trustworthy.

Depth challenge: responses vary; look for perspectives, change over time, and honest unanswered questions.

Lesson 7: Earth's Systems Web

Targets: ESS2-7.6: I can argue from evidence that Earth's systems and life have changed together over time.

Earth's systems web: link cards

Arrows with a star (biosphere changing a non-living system): links 3, 4, 6, 7, 9. Link 4 is an atmosphere effect on water and rock, but the oxygen came from life. Arrows are on the web key. Most arrows pointing out of the biosphere: 3, 6, 7, 9.

Earth's systems web: template and reading chart

Sample reading chart: hot early Earth -> geosphere; oceans form -> hydrosphere; first life -> biosphere; oxygen builds up -> atmosphere (changed by biosphere).

Most arrows out: the biosphere (links 3, 6, 7, 9 and the starred link 4). Most arrows in: the atmosphere or biosphere, depending on the student's extra links. Accept any count that matches the student's web.

Notebook sample answers

Warm-up: geosphere (rock), hydrosphere (water), atmosphere (air), biosphere (living things).

Wrap-up: biosphere has the most arrows out; atmosphere or biosphere the most in.

Lesson 8: Argue Together

Targets: ESS2-7.6: I can argue from evidence that Earth's systems and life have changed together over time.

The timeline: eight rows

Row	How they are linked (sample)
1	Oceans gave a home for first cells (E->L); the cells built mounds in the water (L->E).
2	Cells made oxygen that rusted ocean iron (L->E).
3	Oxygen rose, so cells that use oxygen evolved (E->L).
4	More oxygen and ozone allowed first animals (E->L).
5	Ozone let plants move to land (E->L), and plants made soil (L->E).
6	Forests buried carbon (L->E); high oxygen supported giant insects (E->L).
7	Eruptions changed the climate and ocean oxygen (E->L), killing 96% of ocean species.
8	Asteroid changed climate (E->L); extinctions let mammals diversify (L->E in the living world).

Loops (life changes Earth, then Earth changes life): rows 2 and 3 (oxygen then new cells) or rows 5 and 6. Accept any pair where the effect comes back.

Notebook sample answers

Warm-up: students choose; e.g., the oxygen arrow because it changed everything after it.

Argument: see the timeline key.

Lesson 9: Review and Reteach

Targets: ESS2-7.1: I can name examples of how Earth has changed and how life has changed.; ESS2-7.2: I can describe one way Earth changed life and one way life changed Earth.; ESS2-7.3: I can explain how events in Earth's history changed life, and how life in turn changed Earth.; ESS2-7.4: I can write an argument, with evidence, that Earth and life changed each other.; ESS2-7.5: I can combine several kinds of evidence into one strong argument about Earth and life.; ESS2-7.6: I can argue from evidence that Earth's systems and life have changed together over time.

Review packet A to C

A: 1 photosynthesis; 2 ozone layer; 3 mass extinction; 4 banded iron formation; 5 stromatolite; 6 biosphere, atmosphere; 7 feedback; 8 evidence.

B: ice ages B (or E); photosynthetic cells spread L; Pangaea breaks apart E; land plants spread B (L with effects on soil); oxygen builds up B (L->E); first animals L. Examples vary.

C: 1 L->E (reefs change the shore); waves are slowed and the shore erodes less. 2 E->L; sample chain: ash blocks sunlight -> plants die -> plant-eating animals starve. 3 They take in carbon dioxide / add oxygen; they break rock and add matter to make soil.

Review packet D to F

D: banded iron formation (rock), ancient rock chemistry (data), stromatolite fossils (fossil), computer model (model), story with no data (story). Sample ranking: chemistry and rock formation and fossils strongest (direct and checkable), model (depends on assumptions), story weakest. Combining kinds is stronger because independent evidence that agrees is harder to explain away.

E: Student A = 1 (a claim with no evidence). Student B = 4 (clear claim, evidence from fossil and rock and data with numbers, reasoning, and a counterclaim). Student C = 2 (a claim and some evidence but little reasoning). Rewrite sample: Life changed Earth. Banded iron formations about 2.4 billion years old (rock) and oxygen data rising from about 0% to 15% by 0.5 billion years ago (data) show oxygen increased. Because stromatolite fossils show oxygen-making cells existed, living things must have caused the rise. Someone might say volcanoes made it, but the fossils point to life.

F: Look for a claim that goes both ways, three pieces of evidence from two kinds, one loop (life changes Earth, then Earth changes life), and a counterclaim answered with evidence.

Notebook sample answers

Checklist: students mark honestly; answers vary.

Lesson 10: CSA

Targets: ESS2-7.1: I can name examples of how Earth has changed and how life has changed.; ESS2-7.2: I can describe one way Earth changed life and one way life changed Earth.; ESS2-7.3: I can explain how events in Earth's history changed life, and how life in turn changed Earth.; ESS2-7.4: I can write an argument, with evidence, that Earth and life changed each other.; ESS2-7.5: I can combine several kinds of evidence into one strong argument about Earth and life.; ESS2-7.6: I can argue from evidence that Earth's systems and life have changed together over time.

Notebook sample answers

Reflection: look for banded iron formations as evidence of life changing Earth and for loops (oxygen, soil, extinction).

Matching table: data to source of variation

Use this table to check Lesson 5 and Lesson 6 answers: match each piece of evidence to its kind and what it shows.

Evidence	Kind	What it shows
Banded iron formation	Rock	Oxygen reached the oceans and rusted iron
Stromatolites	Fossil	Photosynthetic life existed 3.5 billion years ago
Ancient rock chemistry (low oxygen before 2.4 billion years ago)	Chemical data	Oxygen was rare, then rose
Computer model of oxygen rise	Model	Photosynthetic spread can raise oxygen
First land plant fossils and ancient soils	Fossil and rock	Plants changed the land and made soil
Mass extinction record	Data	Earth events change life
Reef limestone	Rock	Living things change coasts
"The air was always the same" story	Story	No data; weakest evidence